

COMP.2560: System Programming Assignment #5: Pipes

ASSIGNMENT 4: Overview

The purpose of this assignment is to evaluate your comprehension of Pipes.

There are three (3) questions in this assignment, described below.

ASSIGNMENT 5 – Question 1

The `server.c` and `client.c` pair of code that we discussed in class make use of FIFO and can only transit characters from the client side to the server side.

Modify the source code so that every time the server side receives information from the client side, the server side send an acknowledgement to the client side to display.

For example, the user at the client side types a letter “a”, and the server side displays the letter “a” and then the server side sends the sentence “I received your message.” to the client side so that the client side displays “I received your message.” in its console.

ASSIGNMENT 5 – Question 2

The `game.c` program we discussed in class uses pipes to coordinate the players.

Rewrite the `game.c` program so that it would now use FIFOs to coordinate the players.

ASSIGNMENT 3 – Question 3

Rule #4 for using a pipe says:

4. If a process writes to a pipe whose read-end has been closed, the write operation fails and the writer process receives a `SIGPIPE`.

Write a small program to simulate this scenario as follows.

1. Create a pipe.
2. Close its read-end.
3. Install a signal handler for the `SIGPIPE` signal and this signal handler only prints out a message to say “Caught the `SIGPIPE` signal.” and it returns. This signal handler does not terminate the process.

4. Write to the pipe some information of your choice using the `write(...)` function. For example, an integer, a string, etc.
5. Check the return value of the write function and use `perror(...)` function print out the details of error.

Your code should produce a similar output as the one below:

```
danwu@delta:~/comp2560w2024/labcode$ ./assn5q
caught the sigpipe signal
write function returns -1
pipe problem.: Broken pipe
danwu@delta:~/comp2560w2024/labcode$
```

ASSIGNMENT 4 – Wrap-up and Submission

- For all questions, you need to submit your source code.
Also, use the script command with the timing option to record you compiling and running your program and submit the script file and timing file for each question as well.
Submit the source code for each question and include any accompanying files (if applicable) needed for that question.
Please make sure all your code works properly on CS Linux server.
 - Submit the link to the video you will create.
 - Please properly name all your files so that markers can easily identify which file is for which question.
See the *Assignment Submission Specification document* for naming files in your submissions for this semester.
 - Record one video with audio showing your source code, you compiling and running each program and explaining how each of the underlined items in each question's description is realized in your code.
For the convenience of marking your video submission, please add chapters to your video so that markers can easily jump to the start of each question in your video. If you do not know how to add chapters to your video, please watch this short video to learn how:
 - <https://www.youtube.com/watch?v=d5uSWaCPLj4>
- If you are using Microsoft Streams, this link shows how to add chapters:
- <https://support.microsoft.com/en-us/office/using-manual-chapters-with-videos-on-stream-on-sharepoint-8bbf61eb-c00b-42b5-a514-cce2e45eb6ea>
- Please note the late assignment submission policy stated in the course syllabus.



ASSIGNMENT 5 – Appendix: Grading Rubric

Q.1	- Modify the code so both <code>server.c</code> and <code>client.c</code> can read and write (two-way communication). This involves creating and opening the 2nd <code>FIFO</code> properly: (5 Marks) - Use the 2nd <code>FIFO</code> properly to communicate: (2 Marks) - Print the message successfully on the client side: (1 Mark) Total: 8 Marks
Q.2	- Making the 4 <code>FIFO</code> (1 Mark each): (4 marks) - Open each of the 4 <code>FIFO</code> in correct mode (<code>READ</code> or <code>WRITE</code> mode) (1 Mark each): (4 Marks) - Preserve the functionality of <code>game.c</code> after conversion from pipes to <code>FIFO</code>: (4 Marks) Total: 12 Marks
Q.3	- Create a <code>PIPE</code>: (1 Mark) - Close its read-end: (1 mark) - Install a signal handler for the <code>SIGPIPE</code> signal and this signal handler only prints out a message to say: "Caught the <code>SIGPIPE</code> signal." and it returns. This signal handler does not terminate the process: (2 Marks) - Write to the pipe some information of your choice using the <code>write(...)</code> function. For example, an integer, a string, etc.: (1 Mark) - Check the return value of the write function and use the <code>perror(...)</code> function to print out the details of the error: (1 Mark) - Similar output as the one in the assignment: (2 Marks) Total: 8 Marks
	Submission requirements: 2 Marks
	TOTAL: 30 Marks